Initial Post — Artificial Intelligence (AI) in Banking: Consumer Lending

Link to post: https://www.my-course.co.uk/mod/forum/discuss.php?d=363716#p731580

AI has fundamentally reshaped banking, particularly in credit risk assessment, customer service, and fraud detection (Hentzen *et al*., 2022; Bank of England, 2025). Where early technologies focused on digitisation and rules-based automation, AI marks a further shift toward predictive, data-driven decision-making, enabled by large-scale datasets and increased computing power.

This shift is particularly evident in consumer lending. Historically, banks used human judgement, credit bureau data, rules-based scorecards, and lending policies when assessing the probability of default. These methods often excluded marginalised individuals with limited credit histories, even when their default risk was relatively low. Machine learning systems, including random forests and gradient-boosting methods such as CatBoost, LightGBM, and XGBoost, allow banks to evaluate broader credit-risk data. A recent example is Upstart (2024), which claims its AI lending model improves approval rates for underserved borrowers. However, such claims require caution, because greater access does not necessarily indicate fairness if model outputs remain opaque, proxy variables persist, or risk thresholds continue to disadvantage protected or low-income groups.

These systems also create governance risks. Computing systems are shaped by the cultural and social contexts surrounding their design and implementation (Walton, 2024). In credit risk assessment, this manifests directly: models trained on historical lending data inherit the exclusionary patterns of that history, and apparently neutral variables can act as proxies for race, gender, or ethnicity (Langenbucher and Corcoran, 2022). Kozodoi, Jacob and Lessmann (2022) show that reducing discrimination can entail profitability trade-offs, highlighting the tension between commercial and ethical objectives. Transparency and

explainability are therefore essential, because banks must justify decisions to regulators and borrowers. Explainable AI (XAI) techniques, such as SHapley Additive exPlanations (SHAP), can identify the variables influencing credit decisions, providing interpretable explanations. This aligns with UK regulatory concerns about how AI-augmented lending decisions can be explained to borrowers (Belton *et al*., 2025).

Overall, AI may improve lending efficiency and inclusion, but only if fairness and accountability are embedded in model design, deployment, and oversight.

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Peer Response 1

Link to post: https://www.my-course.co.uk/mod/forum/discuss.php?d=363787#p731908

Your post identifies diagnostic efficiency as a strong use case of AI in healthcare, and McKinney *et al.* (2020) support your argument. Although efficiency and accuracy are key drivers for adopting these technologies, there is a potential trade-off between accuracy and transparency. This raises questions about the usability of AI systems if the reasoning behind diagnoses cannot be easily understood, reflecting the black-box problem. Aravindkumar *et al.* (2026) show that Explainable AI (XAI) can improve transparency and trust in healthcare AI systems, particularly in imaging and diagnostics.

Furthermore, your point about data privacy, while important, leaves open a related question: on whose data are these systems trained? For example, if the training dataset does not contain sufficient racial diversity, there is a fundamental risk that the performance of these systems may vary across different patient groups, complicating the argument that AI screening is more efficient than human screening. Kumar *et al*. (2024) emphasise that healthcare AI is susceptible to data bias, while also showing that issues such as governance, validation, and accountability affect the safe use of such systems. Therefore, while I agree that AI can improve diagnosis in healthcare, its societal benefits depend on transparency, representative data, and appropriate clinical oversight.

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Peer Response 2

Link to post: https://www.my-course.co.uk/mod/forum/discuss.php?d=364010#p733514

Your post makes a strong case that AI is transforming healthcare through cancer diagnostics. I agree that AI detecting more cancers without increasing false positives is important, but I would extend this by considering how screening involves trade-offs between early detection and patient harm.

In screening, accepting some false positives may be clinically justifiable if this helps reduce the risk of false negatives, since missed diagnoses may delay treatment and worsen patient outcomes. The aim of AI screening should be diagnostic optimisation rather than simply maximising detection, which is supported by the Eisemann *et al.* (2025) study. This trade-off is complex because increased detection can also raise the risk of overdiagnosis, meaning that not all screen-detected cancers necessarily improve patient outcomes (Welch and Black, 2010).

Walton's (2024, pp. 149-152) discussion of algorithmic bias raises a further consideration: this trade-off should not be treated as population-neutral. If training data under-represents some demographic groups, average improvements in AI accuracy may mask unequal false-positive or false-negative rates. Therefore, AI should be judged not only by accuracy, but also by its wider clinical consequences. Should successful AI screening, therefore, be defined not only by detection rates, but also by patient outcomes, and by responsible clinical decision-making (i.e., human-in-the-loop oversight)?

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Additional — Peer Response 3

Link to post: https://www.my-course.co.uk/mod/forum/discuss.php?d=364502#p735494

Your post clearly analyses why cybersecurity has become a strategic issue for banking, with particular reference to fraud prevention, operational resilience, and regulatory compliance. I agree that incidents such as the JPMorgan breach show how cybersecurity failures can reshape security investment and industry standards. However, I would extend your argument further by considering how AI is changing the cyber-risk landscape.

The case of Claude Mythos Preview is particularly relevant. This AI system has shown the ability to simulate cyber-attacks, including identifying and exploiting vulnerabilities in controlled test settings (AI Security Institute, 2026). For banks, this suggests that future threats may increasingly involve AI-enabled attackers or AI-assisted tools capable of accelerating vulnerability discovery and exploitation.

Drawing on Walton's (2024) framework, this could be understood as a hybrid socio-technical issue rather than a purely technical one because in banking, cybersecurity affects privacy, consumer trust, and system access. Furthermore, Amonya (2026) frames Claude Mythos as a "narrative engine" which raises concerns about the governance of AI-generated outputs and the difficulty of auditing AI-driven cybersecurity decisions. This supports the view that AI systems require human-in-the-loop validation and robust AI governance frameworks to ensure cybersecurity tools remain accountable, auditable, and transparent.

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Summary Post — Artificial Intelligence (AI) in Banking: Consumer Lending

Link to post: https://www.my-course.co.uk/mod/forum/discuss.php?d=367746#p741315

My initial post argued that AI is transforming banking by reshaping credit risk assessment and potentially expanding lending access to underserved communities. However, I cautioned that higher approval rates do not necessarily mean that AI models, such as those employed by Upstart (2024), are inherently fair. Mariam's feedback strengthened this point by highlighting that financial inclusion claims require closer scrutiny because algorithmic lending may still produce unequal outcomes despite appearing to broaden access (Bartlett *et al*., 2022; Li *et al*., 2024).

Building on this, peer responses shifted the discussion from whether AI expands access to whether that access is fair, understandable, and accountable. They questioned whether AI models can be considered fair when lending decisions are opaque or based on variables that loan applicants do not understand. Although Explainable AI (XAI) tools such as SHAP help banks understand why certain credit decisions are made, banks may be reluctant to share these explanations fully with potential borrowers. This creates a governance tension: explanations are needed for accountability, but excessive disclosure may enable model manipulation or gaming (Ozanne, 2024). Consequently, explainability is not merely a technical issue; it is also a governance problem because banks must manage trade-offs between transparency, fairness, commercial risk, and regulatory accountability.

Benjamin and Oscar emphasised the importance of operational accountability and governance. AI models should augment, rather than replace, human credit decisions. Some XAI tools are applied after credit decisions have been made, meaning they can make a biased model legible without making it fair. If model inputs were inappropriate from the outset, post-hoc explanations cannot correct them (Mehrabi *et al*., 2021). This developed my argument by showing that fairness must be addressed before, during, and

after deployment: through careful feature selection, fairness testing, human oversight, and meaningful post-hoc explanation.

In conclusion, peer feedback and my own responses across consumer lending, healthcare, and cybersecurity strengthened my view that AI is not merely a technical development but a socio-technical governance challenge involving fairness, transparency, and accountability.

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